

Education LMS - Structural & Architectural Blueprint

1. Overall Architecture

Flutter Mobile App (Student, Teacher, Admin) communicates over HTTPS with a single NestJS backend.

The backend manages authentication, RBAC, courses, videos, tests, attendance, notifications and analytics.

PostgreSQL stores data, Cloudflare R2/S3 stores files, Firebase sends push notifications.

2. Structural Layout

Presentation Layer

- Flutter Mobile App
- Next.js Admin Portal

Business Layer

- Authentication
- Users
- Courses
- Lectures
- Tests
- Attendance
- Notifications

Data Layer

- PostgreSQL
- Cloud Storage
- Redis (optional)

3. Application Modules

Student	Teacher	Admin
Dashboard	Dashboard	Analytics
Courses	Create Course	Manage Users
Videos	Upload Videos	Approve Courses
Tests	Create Tests	Reports
Attendance	Attendance	Settings
Notifications	Announcements	Broadcasts
Profile	Profile	RBAC

4. Backend Modules

Auth Module
Users Module
Course Module
Lecture Module
Content Module
Assessment Module
Attendance Module
Notification Module
Analytics Module
Payment Module (optional)

5. Deployment Pipeline

Developer → GitHub → GitHub Actions → Docker → VPS → Nginx → NestJS → PostgreSQL
Flutter → Codemagic → Android/iOS Stores
Next.js Admin → Vercel

6. RBAC Flow

Login → JWT → Role → Permission → Module → API Access
Roles: Super Admin, Admin, Teacher, Student

7. Recommended Folder Structure

apps/
mobile/
admin-web/

backend/
auth/
users/
courses/
lectures/
tests/
attendance/
notifications/

database/
docs/

8. Scaling

Suitable for 100 students using one VPS.
Scale vertically until ~1000 users.
Split into microservices only after significant growth.